Tantalum-on-Silicon Superconducting Qubits: Device Assumptions for a Modular Fault-Tolerant Path

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Correspondence: research@zhyra.xyz

Abstract

This paper examines the tantalum-on-silicon superconducting qubit platform as a candidate device basis for the QONTOS modular architecture. It does not claim that QONTOS has already demonstrated million-qubit device performance. Instead, it evaluates the device assumptions required for the aggressive and stretch architecture scenarios, including millisecond-class coherence, very high two-qubit fidelity, scalable chiplet fabrication, and manageable cryogenic and control integration. The paper treats literature-reported millisecond-class coherence as an encouraging input, then maps that input into realistic questions about logical depth, packaging, calibration, and fabrication yield.

Claim status: literature-informed device feasibility paper with aggressive and stretch targets

Keywords: superconducting qubits, transmon, tantalum, coherence, device roadmap, chiplet fabrication

---

1. Introduction

The QONTOS architecture depends on a superconducting qubit platform that can simultaneously support:

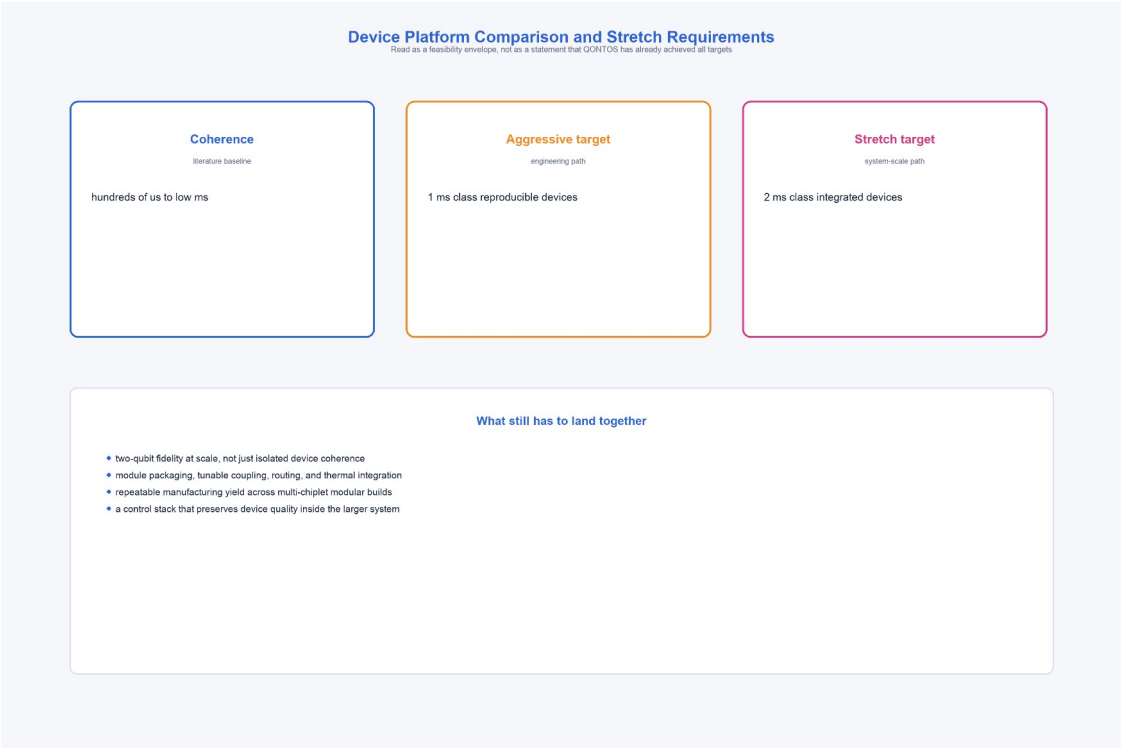
- high coherence
- high gate fidelity
- scalable fabrication
- modular packaging
- practical calibration and control

Tantalum-on-silicon is attractive because literature reports suggest significant progress over earlier material stacks. However, raw coherence values alone do not establish a viable path to fault-tolerant computation. This paper therefore focuses on what device-level performance would actually be needed for the larger architecture to remain credible.

1.1 Claim Status Summary

Claim	Status
Millisecond-class tantalum coherence exists in the literature	Derived from literature
QONTOS can build its stretch architecture on 2 ms devices	QONTOS target / stretch assumption
99.999% two-qubit fidelity at scale is available	Stretch target
Million-qubit fabrication is economically viable	Stretch target

2. Device Performance Requirements



Materials Comparison

Figure 1: Device-platform comparison should be read as a feasibility envelope, not as proof that stretch targets are already achieved.

2.1 Scenario-Based Targets

Metric
T1
T2
1Q fidelity

2Q fidelity

2Q gate time

Readout fidelity

2.2 Literature Baseline

The strongest role of tantalum in the current QONTOS paper set is as a literature-supported direction, not as an already integrated QONTOS device result.

Representative literature-relevant points:

- millisecond-class coherence has been reported for high-quality tantalum devices
- materials quality and oxide behavior appear favorable relative to older stacks
- device integration at scale remains far less mature than isolated coherence headlines

---

3. Coherence, Gate Time, and Logical Depth

3.1 Raw Operations Are Not Logical Depth

The most important correction in this paper is conceptual:

Raw coherence time does not equal executable logical algorithm depth.

For example:

```
Raw physical gate opportunities = T1 / gate_time

If T1 = 2 ms and 2Q gate time = 25 ns:
raw physical gate opportunities = 2,000,000 ns / 25 ns = 80,000
```

That number is not the same thing as:

- logical gate depth
- full algorithm depth
- fault-tolerant executable workload length

In practice, logical depth depends on:

- repeated error-correction cycles
- decoder latency
- leakage handling
- calibration stability
- state preparation and measurement overhead
- idling and synchronization across modules

3.2 Implication for QONTOS

The relevant device question is therefore not:

```
How many raw gates fit inside T1?
```

It is:

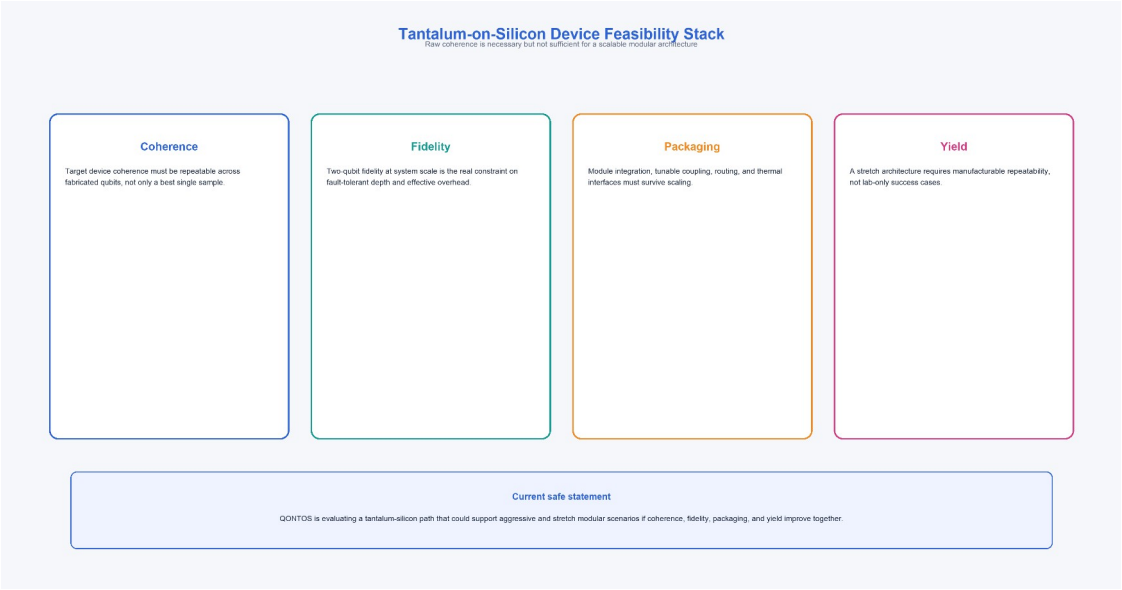
```
Does this device regime reduce the burden on fault tolerance enough
for the system-level architecture to remain plausible?
```

That is the right role for the tantalum platform in the QONTOS story.

3.3 Workload-Oriented Device Requirements

Workload class	Device implication
Early logical-qubit experiments	aggressive case may be sufficient
Modular fault-tolerance demonstrations	requires aggressive to stretch device regime
FeMoco-class flagship workloads	requires stretch device regime plus strong QEC and interconnect assumptions

4. Device Architecture and Control



Device Feasibility Stack

Figure 2: Raw coherence matters, but scalable device feasibility depends on fidelity, packaging, and yield together.

4.1 Double-Transmon Coupler Targets

Parameter
Tunable coupling window

CZ gate fidelity

CZ gate time

ZZ suppression

4.2 What Must Be Validated

The coupler architecture must be evaluated not only for isolated gate performance, but for:

- frequency crowding
- calibration drift
- cross-talk at dense packing
- leakage under repeated operation
- compatibility with chiplet packaging

---

5. Fabrication and Packaging Strategy

5.1 Chiplet Assumptions

The current canonical stretch architecture assumes:

- 2,000 physical qubits per chiplet
- 5 chiplets per module
- 10,000 physical qubits per module

These are architecture constants, not demonstrated manufacturing outputs.

5.2 Fabrication Questions That Matter More Than Headlines

Question	Why it matters
Can yield remain acceptable at chiplet scale?	stretch architecture fails if yield collapses
Can packaging preserve qubit performance?	good bare-die metrics do not guarantee module-level success
Can control density remain manageable?	wiring and control are often harder than the qubits themselves
Can calibration labor be reduced enough?	manual tuning does not scale to large chiplet fleets

5.3 Partner Strategy

Named fabrication partners should be described as a partner strategy or target ecosystem, not as evidence that million-qubit production capacity has already been secured.

Recommended phrasing:

```
QONTOS is exploring a multi-partner fabrication strategy to reduce
supply-chain concentration and improve scale flexibility.
```

6. Thermal and Integration Constraints

6.1 Module Thermal Envelope

The current stretch planning envelope assumes roughly:

Component
Coaxial lines
Optical fibers
DC lines
Chip dissipation
Total
Available budget

This should be presented as a planning estimate tied to the cryogenic paper, not as an experimentally validated module budget.

6.2 Packaging and Calibration Risks

The device story is not only about physics. It is equally about:

- packaging yield
- thermal interfaces
- connector density
- calibration automation
- module replacement and maintainability

If these remain unresolved, even excellent single-device metrics will not convert into a viable modular machine.

7. Device Roadmap by Scenario

Year	Base	Aggressive	Stretch
2025-2026	literature benchmarking, design studies, calibration tooling	small internal validation targets	first evidence supporting millisecond-class stretch path
2026-2027	device-package integration	module-scale integration targets	10,000 physical-qubit stretch module assumptions tested
2027-2028	robust modular	high-fidelity chiplet	architecture-scale

	hardware support	fleet	stretch device envelope tested
2029-2030	strong device basis for modular products	large modular fault-tolerance path	full stretch device assumptions required

8. Validation Gates

The materials and device story should not be considered strong enough for the stretch architecture until the following are satisfied:

1. coherence and fidelity are shown under realistic packaging conditions
2. chiplet-level control and calibration complexity are bounded
3. fabrication yield assumptions are grounded in a real partner process model
4. module thermal and control density assumptions survive systems review

---

9. Conclusion

Tantalum-on-silicon remains a promising candidate platform for the QONTOS modular architecture, especially because it offers a credible path toward stronger coherence and better device quality than older superconducting stacks. But this paper should not present raw coherence as equivalent to fault-tolerant algorithm readiness.

The technically defensible conclusion is:

- literature results make tantalum worth pursuing
- aggressive modular hardware scenarios may be plausible if packaging and control are solved
- the full million-qubit stretch architecture requires substantial additional validation beyond device-level headline metrics

---

References

[1] Representative literature on tantalum transmons and millisecond-class coherence.

[2] Representative literature on superconducting gate fidelity, packaging, and calibration limits.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Device-feasibility analysis, not demonstrated stretch performance